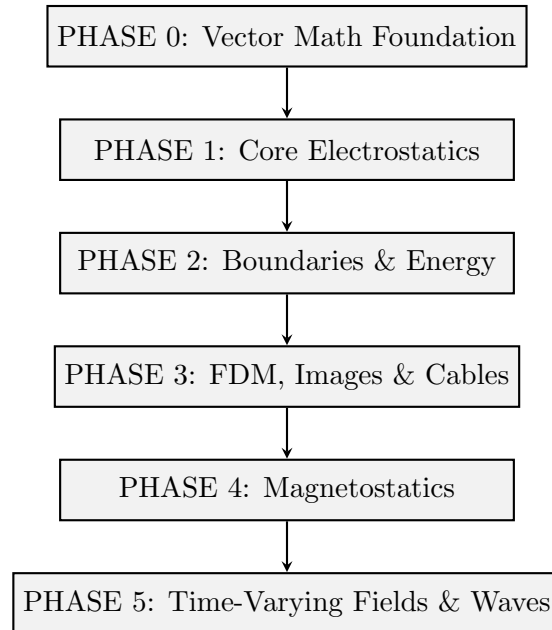


Field Theory: The Ultimate Roadmap

Semester Domination Guide: 0 to 100

Based on 2022-2025 PYQs & Master Notes



PHASE 0: Vector Math Foundation (Weeks 1-2)

Objective: Master the mathematical operators required for the rest of the course.

Core Topics:

- **Coordinate Systems:** Cartesian, Cylindrical, Spherical. Identify constant-coordinate surfaces.
- **The Del Operator (∇):** Gradient, Divergence, and Curl.
- **Theorems:** Divergence Theorem & Stokes' Theorem statements.
- **Derivations (High Yield):** Deriving Divergence in Cylindrical coordinates and Curl in Cartesian coordinates from first principles.

Mandatory Diagrams to Practice:

- **Spherical Surfaces:** A 3D drawing showing a Sphere ($r = c$), Cone ($\theta = c$), and Half-Plane ($\phi = c$) intersecting.
- **Divergence Volume Element:** A differential cylindrical wedge ($dr, r d\phi, dz$).

- **Curl Area Element:** A differential Cartesian rectangle (dy, dz) showing circulation paths.

PYQ Checklist:

- ☐ Derive Divergence in Cylindrical coords (2024).
 - ☐ Derive Curl in Cartesian coords (2022, 2025).
 - ☐ State/Establish Stokes' Theorem (2023, 2024, 2025).
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PHASE 1: Core Electrostatics (Weeks 3-4)

Objective: Understand how stationary charges create fields and potentials.

Core Topics:

- **Coulomb's Law & Superposition:** Calculating $\mathbf{E}$ for discrete charges (e.g., right isosceles triangles).
- **Continuous Charge Distributions:** Deriving $\mathbf{E}$ on the axis of a uniform Ring and a Disc, and equating them.
- **Gauss's Law:** Integral form and proving the Differential form ($\nabla \cdot \mathbf{D} = \rho_v$).
- **Equipotential Surfaces:** Proving orthogonality to electric flux lines.

Mandatory Diagrams to Practice:

- **Ring & Disc Superposition:** A 3D axis showing a disc in the y-z plane and a ring shifted on the x-axis.

PYQ Checklist:

- ☐ Right isosceles triangle superposition (2023).
 - ☐ Ring and Disc superposition (2024, 2025).
 - ☐ Prove $\nabla \cdot \mathbf{D} = \rho_v$ (2022, 2025).
 - ☐ Prove flux lines are perpendicular to equipotentials (2023).
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PHASE 2: Boundaries, Energy, & Conductors (Week 5)

Objective: Learn what happens to fields at material interfaces.

Core Topics:

- **Boundary Conditions (Dielectric):** Tangential $\mathbf{E}$ is continuous ($E_{t1} = E_{t2}$); Normal $\mathbf{D}$ is discontinuous ($D_{n1} - D_{n2} = \rho_s$).
- **Conductor-Dielectric Boundaries:** Field intensity "on" vs "off" the surface ($\sigma/2\epsilon$ vs σ/ϵ).
- **Electrostatic Energy:** Deriving $w_e = \frac{1}{2}\epsilon E^2$ from assembling charges or mechanical pressure.

PYQ Checklist:

- ☐ Derive $E_{t1} = E_{t2}$ and $D_{1n} - D_{2n} = \rho_s$ (2022, 2023, 2024).
 - ☐ Derive energy density $w_e = \frac{1}{2}\epsilon E^2$ (2022, 2023).
 - ☐ Explain field intensity "on" vs "off" conductor surface (2022, 2024).
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PHASE 3: FDM, Images, & Cables (Weeks 6-7)

Objective: High-weight numerical and applied problems.

Core Topics:

- **Uniqueness Theorem:** Statement and proof.
- **Numerical Methods (FDM):** Deriving the 5-point Finite Difference Method formulas using Taylor series (equal and unequal nodal spacing). Solving grids.
- **Method of Images:** Point charge over grounded plane (calculate stresses). Infinite line charge (prove equipotentials are cylinders).
- **Cables:** Field Utilization Factor (η). Capacitance grading. Condition for max stress not occurring at the core ($\epsilon_1 r > \epsilon_2 r_1$).

Mandatory Diagrams to Practice:

- **FDM Stencils:** A cross-shaped grid showing nodes 0, 1, 2, 3, 4.
- **Method of Images:** Geometry of a point charge $+Q$, its image $-Q$, and the observation point P .
- **Cable Cross-section:** Concentric circles representing core and dielectric layers.

PYQ Checklist:

- ☐ State and prove Uniqueness Theorem (2022, 2025).
- ☐ FDM derivation and grid solving (2022, 2024).
- ☐ Method of images: Point charge stresses (2022, 2023).
- ☐ Method of images: Line charge cylinders proof (2022, 2024).

- ☐ Cable capacitance grading / Utilization factor (2022, 2024).
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PHASE 4: Steady Magnetic Fields (Week 8)

Objective: Steady currents and magnetic fields.

Core Topics:

- **Continuity Equation:** Prove $\nabla \cdot \mathbf{J} = 0$ for steady currents.
- **Ampere's Law:** Establish $\nabla \times \mathbf{H} = \mathbf{J}$.
- **Vector Potential ($\mathbf{A}$):** Deduce Biot-Savart Law from $\mathbf{A}$.
- **Biot-Savart Applications:** Calculate $\mathbf{H}$ or $\mathbf{B}$ on the axis of a square/rectangular current loop.
- **Magnetic Boundary Conditions:** Derivation and application.

PYQ Checklist:

- ☐ Establish $\nabla \times \mathbf{H} = \mathbf{J}$ (2023, 2024, 2025).
 - ☐ Square/Rectangular loop B-field calculation (2023, 2024).
 - ☐ Divergence of $\mathbf{J}$ equals 0 (2023, 2025).
 - ☐ Vector potential & Biot-Savart deduction (2022).
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PHASE 5: Time-Varying Fields & EM Waves (Weeks 9-10)

Objective: Maxwell's equations and wave propagation.

Core Topics:

- **Faraday's Law:** Establish $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ with a pictorial view.
- **Displacement Current:** Establish $\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t$ with a pictorial view.
- **Poynting Theorem:** Derivation of $\mathbf{P} = \mathbf{E} \times \mathbf{H}$.
- **EM Waves:** Derive wave equations in free space and solve analytically for plane waves.

Mandatory Diagrams to Practice:

- **Displacement Current:** Capacitor circuit showing flat surface S_1 and balloon surface S_2 .
- **Faraday's Law:** Increasing $\mathbf{B}$ field through a loop inducing $\mathbf{E}$.
- **Plane Wave:** Orthogonal sine waves of $\mathbf{E}$ and $\mathbf{H}$ propagating along the z-axis.

PYQ Checklist:

- ☐ Derive $\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t$ & pictorial view (2022, 2024, 2025).
- ☐ Derive $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ & pictorial view (2022, 2023, 2024, 2025).
- ☐ Poynting Theorem derivation (2024, 2025).
- ☐ Wave equations in free space & plane wave analytical solution (2023, 2024, 2025).